

3 Ans: D

Find speed v just before hitting the ground:

$$v^2 = u^2 + 2as$$

$$v^2 = 0 + 2(9.81)(3.00)$$

$$v = 7.67 \text{ m s}^{-1}$$

Find resultant force F :

$$F = \frac{\Delta p}{\Delta t}$$

$$F = \frac{(80.0)(7.67 - 0)}{0.0250} = 24550 \text{ N}$$

Average force – weight = resultant force

Average force by surface on mass – $mg = ma$

Average force by surface on mass = $ma + mg = 24550 + (80.0)(9.81) = 25300 \text{ N}$